# Entity resolution, step by step

This notebook runs the pipeline that `scripts/resolve.py` runs on a file of any size: the same `er.run.Run`, one step per section, on a small generated population (about 3,700 records) so that every step can be seen. It holds no logic of its own; each section calls the code, then charts what the step did. Every code cell prints its Big-O, filled in with this file's numbers, and its time. The whole notebook runs in about half a minute.

The run never reads the truth (which records are one entity). It exists only because the data is generated, and the notebook reads it to show what each step got right.

1 Setup · 2 Generate · 3 Standardize · 4 Fit · 5 Rules · 6 Candidates · 7 Score · 8 Why a pair matches · 9 Threshold · 10 Cluster and evaluate · 11 Increment · 12 How the configuration was chosen · 13 Cost as the file grows

N records, L characters per value, P pairs inside a key's blocks, C candidate pairs, K comparisons, S sampled pairs.

## 1 · Setup

The chosen pipeline is `config/final.toml`, which `scripts/choose.py` wrote. The demo's population is `config/generator.toml`'s, made smaller. Charts use one colour per meaning: blue for the same entity (or evidence for a match), orange for different entities, red for evidence against.

In [1]:
```
import logging
import math
import shutil
import time
from contextlib import contextmanager
from dataclasses import replace

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from matplotlib.ticker import FuncFormatter, PercentFormatter
```

```
from matplotlib_inline.backend_inline import set_matplotlib_formats

from er.candidates import found_by
from er.config import column_name, load_pipeline
from er.evaluate import cluster_metrics, missed_pairs, pair_metrics, predicted_pairs
from er.generate import check, generate, load_reference, load_spec
from er.io import table_after
from er.match import parameters, scaled_prior, waterfall
from er.resolve import connected_components
from er.run import Run, fill_rates
from er.settings import load_settings

logging.disable(logging.WARNING)
S = load_settings()
FINAL = load_pipeline(S.root / "config" / "final.toml", S.standardize, S.chains, S.columns)
SPEC = replace(load_spec(S.root / "config" / "generator.toml")[0], entities=3_000)  # the demo's size
DEMO = S.paths["artifacts"] / "demo"
DEMO.mkdir(parents=True, exist_ok=True)
UID, ENT = S.data.unique_id, S.data.entity_id
L, R = f"{UID}_l", f"{UID}_r"
FIELDS = [f for f in S.columns if f not in ("id", "entity")]  # the logical fields of config/settings.toml [column
s]
pd.set_option("display.max_columns", 30, "display.width", 200)

BLUE, ORANGE, RED, INK, INK2, MUTED, GRID, AXIS, SURFACE = ("#2a78d6", "#eb6834", "#e34948", "#0b0b0b", "#52514e",
                                                            "#898781", "#e1e0d9", "#c3c2b7", "#fcfcfb")
set_matplotlib_formats("svg")
plt.rcParams.update({"figure.facecolor": SURFACE, "axes.facecolor": SURFACE, "savefig.facecolor": SURFACE,
                     "axes.edgecolor": AXIS, "axes.linewidth": 0.8, "axes.labelcolor": INK2, "axes.titlecolor": IN
K,
                     "axes.titlesize": 10, "axes.titleweight": "bold", "axes.titlelocation": "left",
                     "axes.spines.top": False, "axes.spines.right": False, "axes.grid": True, "axes.axisbelow": Tr
ue,
                     "grid.color": GRID, "grid.linewidth": 0.6, "xtick.color": MUTED, "ytick.color": MUTED,
                     "xtick.labelcolor": INK2, "ytick.labelcolor": INK2, "font.size": 9, "lines.linewidth": 2,
                     "legend.frameon": False, "figure.dpi": 100})


@contextmanager
def timed(big_o: str):  # a cell's own work, outside the run's steps
    t = time.perf_counter()
```

```
    yield
    print(f"{big_o} · {time.perf_counter() - t:.2f}s")


def done(run: Run) -> None:  # the step just run: its Big-O with this file's numbers, and its time
    s = run.steps[-1]
    print(f"{s['step']}: {s['big_o']} = {s['filled']} · {s['seconds']:.2f}s")


def same(pairs: pd.DataFrame) -> np.ndarray:  # the truth, read only to show what a step got right
    return pairs[L].map(entity).to_numpy() == pairs[R].map(entity).to_numpy()


def finish(fig) -> None:
    fig.tight_layout()
    plt.show()
```

## 2 · Generate

A synthetic population built from public reference data: each entity, a person or a company, is written as one or more records with typos, empty fields and life changes (a move, a new phone, a married surname). The truth is kept apart. Below: the entity whose three records differ the most, then how many records each entity has and how often each column is filled. O(E·r) for E entities of r records each.

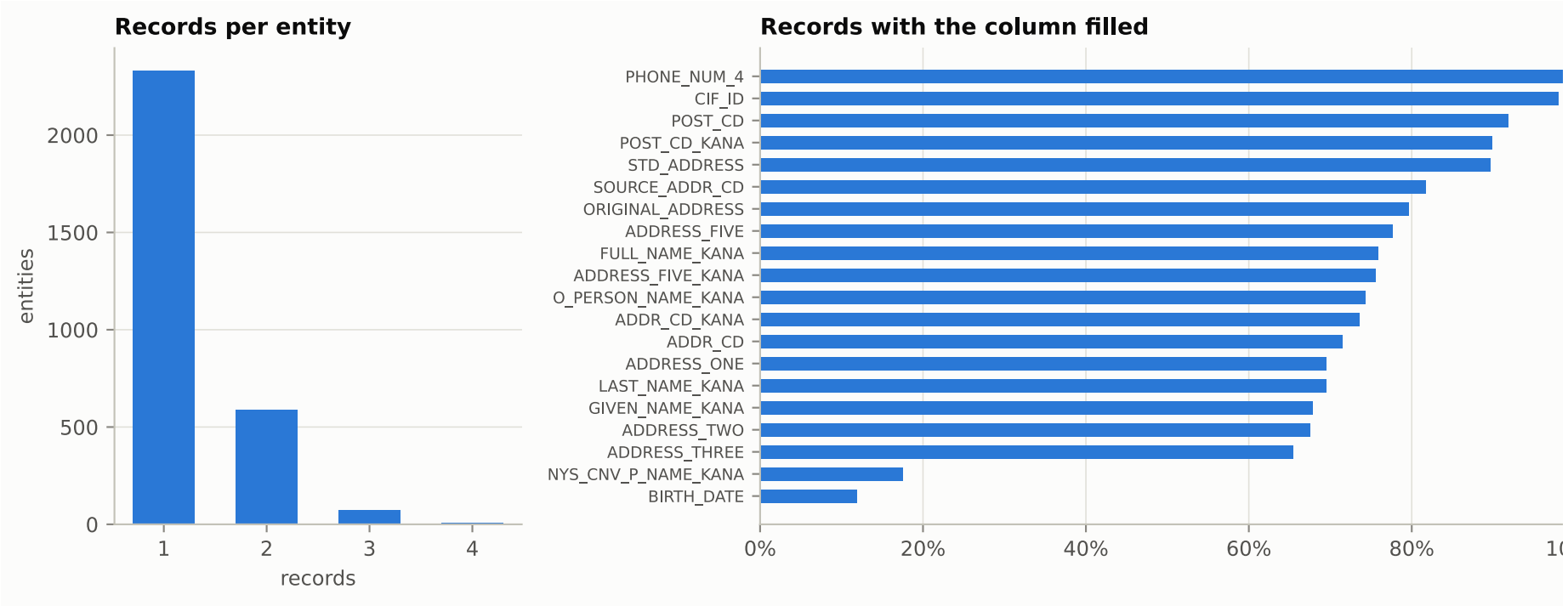
In [2]:
```
with timed(f"generate: O(E·r) = O({SPEC.entities:,} entities · their records)"):
    g = generate(SPEC, *load_reference(S.paths["reference"]))
    check(SPEC, g)
g.df.to_parquet(DEMO / "records.parquet", index=False)
entity = g.truth.set_index(UID)[ENT]
per_entity = entity.value_counts()
print(f"{len(g.df):,} records · {len(per_entity):,} entities · {(per_entity > 1).mean():.0%} with more than one re
cord")

raw = g.df.set_index(UID)
varied = raw.groupby(entity.reindex(raw.index).to_numpy()).nunique().gt(1).sum(axis=1)  # columns that differ, per
entity
example = varied[per_entity.reindex(varied.index).eq(3)].idxmax()
ex_ids = entity.index[entity == example]
display(raw.loc[ex_ids].dropna(axis=1, how="all"))
```

```
fig, (a, b) = plt.subplots(1, 2, figsize=(10, 3.8), width_ratios=[1, 2])
counts = per_entity.value_counts().sort_index()
a.bar(counts.index.astype(str), counts.to_numpy(), width=0.6, color=BLUE)
a.set(title="Records per entity", xlabel="records", ylabel="entities")
a.grid(axis="x", visible=False)
fill = raw.notna().mean().sort_values()
b.barh(fill.index, fill.to_numpy(), height=0.6, color=BLUE)
b.set(title="Records with the column filled", xlim=(0, 1))
b.grid(axis="y", visible=False)
b.xaxis.set_major_formatter(PercentFormatter(1))
b.tick_params(axis="y", labelsize=7)
finish(fig)
```

```
generate: O(E·r) = O(3,000 entities · their records) · 0.77s
3,758 records · 3,000 entities · 22% with more than one record
```

| CONT_ID | PHONE_NUM_4 | CIF_ID | FULL_NAME_KANA | O_PERSON_NAME_KANA | POST_CD | POST_CD_KANA | LAST_NAME_KANA | GIVEN_NAME_KANA | BIRTH_DATE | SOURCE_ADDR_CD | ORIGINAL_ADDRESS | ADDRESS_FIVE | ADDRESS_FIVE_KANA | ADDR_CD_KANA | ADDR_CD | STD_ADDRESS | ADDRESS_ONE | ADDRESS_TWO | ADDRESS_THREE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0000000536 | 4573 | 6243089361260 | ｸﾎﾞ ｳﾝ | ｸﾎﾞ ｳﾝ | 8160831 | 8160831 | ｸﾎﾞ | ｳﾝ | 19350813 | 9035229242 | 春日市大谷９－１８－９ | 9-18-9 | 9-18-9 | 40218009009 | 40218009009 | 福岡県春日市大谷9-18-9 | 福岡県 | 春日市 | 大谷 |
| 0000001247 | 9503 | 8517481989993 | ｸﾎﾞｵﾝ | ｸﾎﾞ ｵﾝ | 8000301 | 8000301 | ｸﾎﾞ | ｵﾝ | 19360813 | 8037260253 | 福岡県京都郡苅田町松原町5-27グランドール602 | 5-27 | 5-27 | 40621003000 | 40621003000 | 福岡県京都郡苅田町松原町5-27 | 福岡県 | 京都郡苅田町 | 松原町 |
| 0000001654 | 4573 | 5633639874782 | クボウン | クボ ウン | 8160831 | 8160831 | クボ | ウン | <NA> | 9035229242 | <NA> | <NA> | <NA> | 40218009009 | <NA> | <NA> | <NA> | <NA> | <NA> |




## 3 · Standardize

`load` reads the file in chunks and checks it against `config/pipeline.toml` `[fields]`: the id is mandatory, and at least one of the person and company names is needed. An optional field the file lacks disables only what reads it. It then builds each logical field from `config/settings.toml` `[columns]` (the first filled source wins) and the normalizations the pipeline reads: kana folding, legal forms dropped, addresses cut to the street number. Each record is standardized on its own, so chunks change nothing. O(N·L). Last, the checks `scripts/resolve.py --check` prints: each field's fill rate, the share of filled values that look like the field ( `[locale.detect]` ), and the share of its most common value, where a placeholder such as `0000` would show.

In [3]:
```
run = Run(FINAL, S)
run.load(DEMO / "records.parquet")
done(run)
print("disabled:", ", ".join(run.disabled) or "nothing: every optional field is in the file")
std = run.con.sql("SELECT * FROM records").df().set_index(UID)
```

```
display(std.loc[ex_ids, FIELDS])  # the entity above, in logical fields
rows = []
for field, chain in [("person_name", "fold"), ("company_name", "company"), ("address", "address")]:
    after = column_name(field, S.chains[chain])
    changed = std.loc[std[field].notna() & (std[field] != std[after]), [field, after]].head(3)
    rows.append(changed.set_axis(["before", "after"], axis=1).assign(field=field, normalization=" → ".join(S.chain
s[chain])))
display(pd.concat(rows)[["field", "normalization", "before", "after"]])
display(fill_rates(run, FIELDS).set_index("field").round(3))
```

```
load: O(N·L) = O(3,758·L) · 0.09s
disabled: nothing: every optional field is in the file
```

| CONT_ID | full_name | person_name | given_name | company_name | any_name | address | address_code | postal_code | prefecture | town_code | city | phone | customer_id | birth_date |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0000000536 | クボウン | クボウン | ウン | NaN | クホウン | 福岡県春日市大谷9-18-9 | 9035229242 | 8160831 | 福岡県 | 40218009009 | 春日市 | 4573 | 6243089361260 | 19350813 |
| 0000001247 | クボオン | クボオン | オン | NaN | クホオン | 福岡県京都郡苅田町松原町5-27 | 8037260253 | 8000301 | 福岡県 | 40621003000 | 京都郡苅田町 | 9503 | 8517481989993 | 19360813 |
| 0000001654 | クボウン | クボウン | ウン | NaN | クホウン | NaN | 9035229242 | 8160831 | NaN | NaN | NaN | 4573 | 5633639874782 | NaN |

| CONT_ID | field | normalization | before | after |
|---|---|---|---|---|
| 0000000005 | person_name | kana_fold | タジツコーダイ | タシツコウタイ |
| 0000000006 | person_name | kana_fold | マツザキトモアギ | マツサキトモアキ |
| 0000000008 | person_name | kana_fold | ヨネクラギョーシン | ヨネクラキヨウシン |
| 0000000007 | company_name | nfkc → strip_legal_form → kana_fold | イスズエンジンセイゾウホッカイドウカブシキガイシャ | イススエンシンセイソウホツカイトウ |
| 0000000013 | company_name | nfkc → strip_legal_form → kana_fold | カブシキガイシャグラニツト | クラニツト |
| 0000000017 | company_name | nfkc → strip_legal_form → kana_fold | ナルミヤインターナショナル | ナルミヤインタウナシヨナル |
| 0000000002 | address | address | 神奈川県海老名市門沢橋2-23-5 | 海老名市門沢橋2-23-5 |
| 0000000003 | address | address | 福島県福島市北町5-30 | 福島市北町5-30 |
| 0000000004 | address | address | 佐賀県佐賀市12-26 | 佐賀市12-26 |

| field | filled | looks_right | top_value | flag |
|---|---|---|---|---|
| full_name | 0.759 | 1.000 | 0.001 | |
| person_name | 0.779 | 1.000 | 0.001 | |
| given_name | 0.679 | 1.000 | 0.004 | |
| company_name | 0.175 | 0.798 | 0.003 | |
| any_name | 0.954 | 1.000 | 0.001 | |
| address | 0.936 | 0.788 | 0.003 | |
| address_code | 0.817 | 1.000 | 0.002 | |
| postal_code | 0.928 | 1.000 | 0.002 | |
| prefecture | 0.695 | 1.000 | 0.115 | |
| town_code | 0.715 | 1.000 | 0.003 | |
| city | 0.675 | 1.000 | 0.009 | |
| phone | 1.000 | 1.000 | 0.001 | |
| customer_id | 0.980 | 1.000 | 0.001 | |
| birth_date | 0.119 | 0.962 | 0.004 | |

## 4 · Fit, without labels

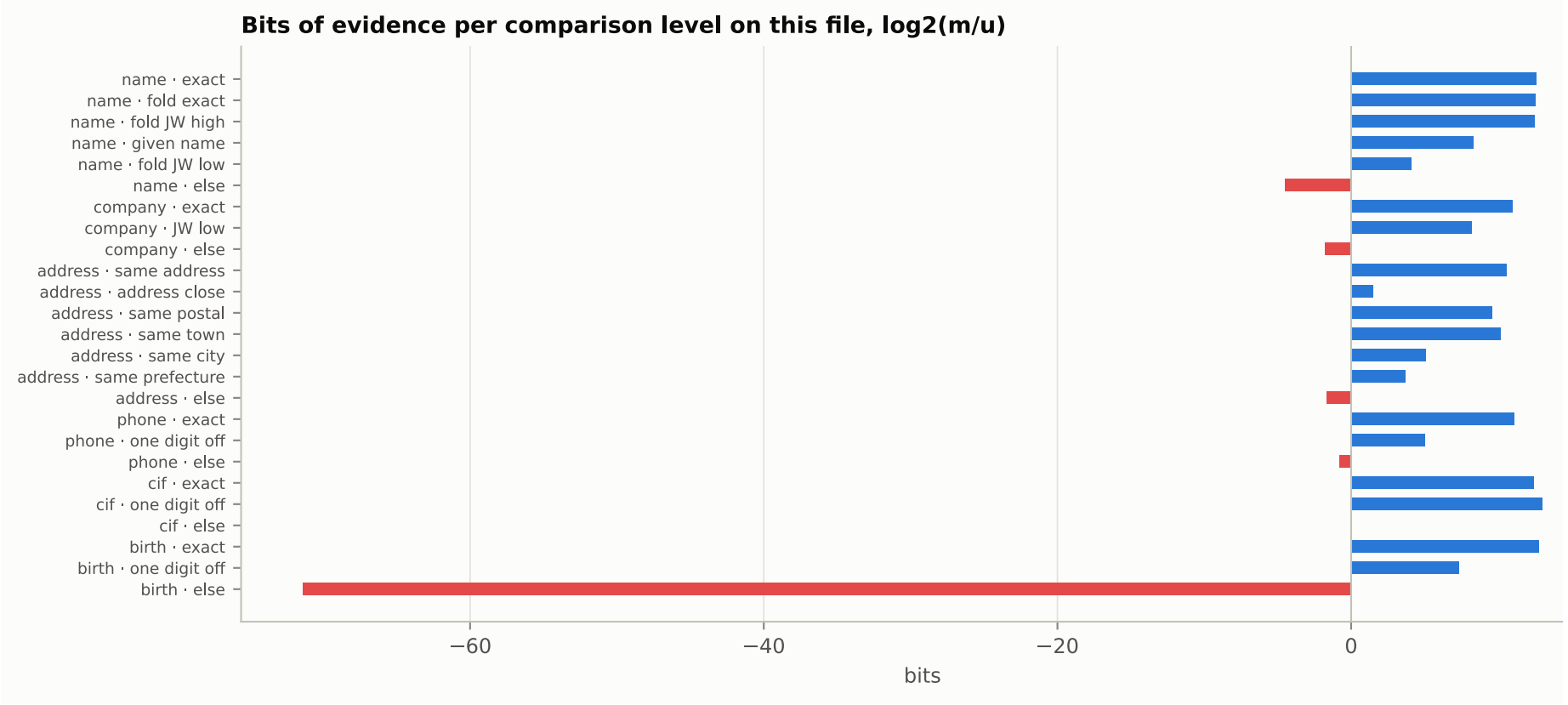
Fellegi–Sunter weighs each comparison's agreement level by how much likelier it is for a true match than by chance: log2(m/u) bits. With no labels, λ (the chance two random records match) comes from near-certain rules and their measured recall; u (agreement by chance) from value counts for each comparison's leading exact levels and a random sample of pairs for the rest; m (agreement for true matches) by EM on blocks of equal phones or addresses. The fit uses a fixed-size sample of whole postal areas, so its cost doesn't grow with the file. Then u is measured again on the whole file: how often two records agree by chance depends on how many records there are, and u from the sample would understate the evidence of an agreement on the file. The gate on oversized blocks (section 6) and the scores (section 7) both use these. Below: the bits each level is worth on this file. O(n + S·K + P_em·K) + O(N + S·K).

In [4]:
```
model = run.fit()
done(run)
lam = scaled_prior(model, run.n)
print(f"λ = {lam:.2e} at {run.n:,} records: a pair starts at {math.log2(lam / (1 - lam)):.1f} bits")
params = parameters(run.scoring).dropna(subset=["weight"]).iloc[::-1]

fig, ax = plt.subplots(figsize=(10, 4.4))
ax.barh(params["comparison"] + " · " + params["level"], params["weight"], height=0.6,
        color=np.where(params["weight"] >= 0, BLUE, RED))
```

```
ax.axvline(0, color=AXIS, linewidth=0.8)
ax.set(title="Bits of evidence per comparison level on this file, log2(m/u)", xlabel="bits")
ax.grid(axis="y", visible=False)
ax.tick_params(axis="y", labelsize=7)
finish(fig)
```

```
fit: O(n + S·K + P_em·K) + O(N + S·K) = n = 3,758; u on 3,758 records · 5.38s
λ = 1.47e-04 at 3,758 records: a pair starts at −12.7 bits
```




## 5 · Rules

Stage 1 is deterministic. An exact rule joins records equal on all its keys; a fuzzy rule needs some keys equal and others close (names within two edits, company names by shared characters, addresses by Jaro–Winkler, a phone one digit off). Before any join, `budget` counts every rule's pairs P = Σ C(k, 2) over its key blocks and stops the run if one is over budget. The first rule in order wins a pair, and a

fuzzy pair is kept only if it touches a record the exact rules left alone. Left: how many true pairs the rules join, clusters included. Right: the true pairs each rule joins itself.

In [5]:
```
plan = run.budget()
done(run)
run.rules()
done(run)
kept = run.con.sql("SELECT * FROM rule_kept").df().assign(true=same)
order = [r.name for st in FINAL.rules.stages for r in st.rules]
stage_of = {r.name: st.name for st in FINAL.rules.stages for r in st.rules}
per_rule = (plan[plan["kind"] == "rule"].set_index("name")[["est_pairs", "largest_block"]].rename(columns={"est_pa
irs": "P"})
            .join(kept.groupby("rule")["true"].agg(joined="size", true="sum")).fillna(0).astype(int).reindex(orde
r))
display(per_rule.rename_axis("rule"))

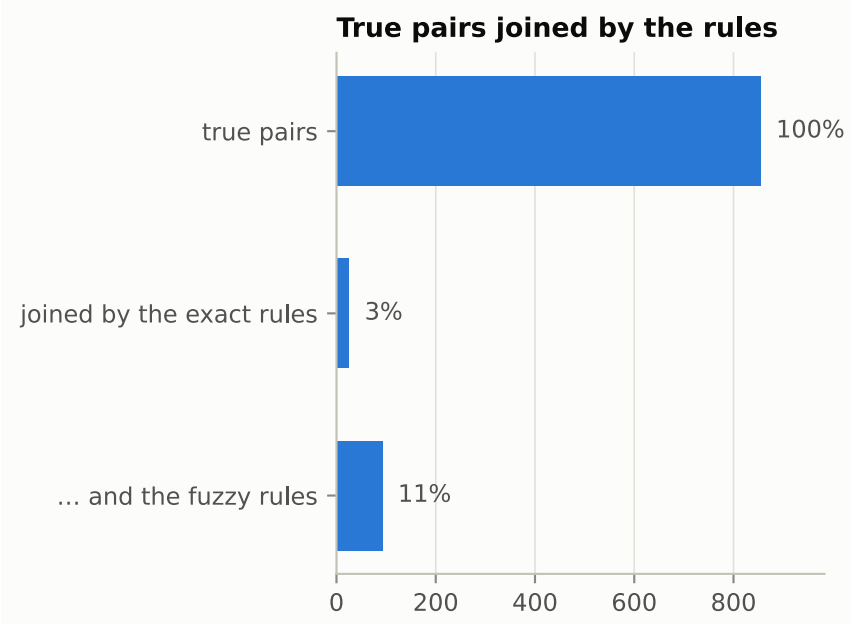
rule_clusters = run.con.sql("SELECT * FROM rule_clusters").df().set_index(UID)["cluster_id"]
exact_only = connected_components(pd.Series(std.index), kept[kept["stage"] == "exact"], run.db, UID)
m_exact, m_rules = (pair_metrics(c, entity.reindex(c.index)) for c in (exact_only, rule_clusters))
total = m_rules["TP"] + m_rules["FN"]
print(f"the rules join {m_rules['recall']:.1%} of the {total:,} true pairs, with {m_rules['FP']} false")

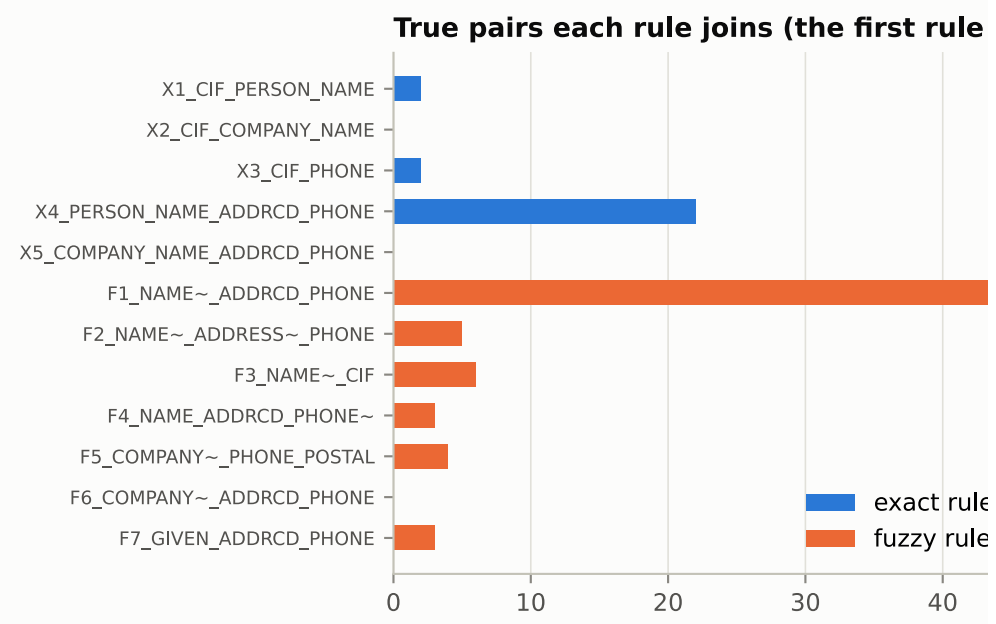
fig, (a, b) = plt.subplots(1, 2, figsize=(10, 3.4), width_ratios=[1, 1.3])
funnel = pd.Series({"true pairs": total, "joined by the exact rules": m_exact["TP"], "… and the fuzzy rules": m_ru
les["TP"]})[::-1]
a.barh(funnel.index, funnel.to_numpy(), height=0.6, color=BLUE)
for y, v in enumerate(funnel.to_numpy()):
    a.text(v, y, f"  {v / total:.0%}", va="center", color=INK2)
a.set(title="True pairs joined by the rules", xlim=(0, total * 1.15))
a.grid(axis="y", visible=False)
y = np.arange(len(order))[::-1]
for stage, colour in (("exact", BLUE), ("fuzzy", ORANGE)):
    mask = np.array([stage_of[r] == stage for r in order])
    b.barh(y[mask], per_rule["true"].to_numpy()[mask], height=0.6, color=colour, label=f"{stage} rules")
b.set_yticks(y, order, fontsize=7)
b.set(title="True pairs each rule joins (the first rule wins a pair)")
b.grid(axis="y", visible=False)
b.legend(loc="lower right")
finish(fig)
```

budget: O(N) = O(3,758) per rule and generator · 0.06s
rules: O(N + Σ P_r·L²) + O(E·d + N) = O(3,758 + 91 pairs) · 0.04s

| rule | P | largest_block | joined | true |
|---|---|---|---|---|
| X1_CIF_PERSON_NAME | 2 | 2 | 2 | 2 |
| X2_CIF_COMPANY_NAME | 0 | 1 | 0 | 0 |
| X3_CIF_PHONE | 3 | 2 | 2 | 2 |
| X4_PERSON_NAME_ADDRCD_PHONE | 23 | 2 | 22 | 22 |
| X5_COMPANY_NAME_ADDRCD_PHONE | 0 | 1 | 0 | 0 |
| F1_NAME~_ADDRCD_PHONE | 89 | 2 | 44 | 44 |
| F2_NAME~_ADDRESS~_PHONE | 83 | 3 | 5 | 5 |
| F3_NAME~_CIF | 14 | 2 | 6 | 6 |
| F4_NAME_ADDRCD_PHONE~ | 58 | 3 | 3 | 3 |
| F5_COMPANY~_PHONE_POSTAL | 107 | 3 | 4 | 4 |
| F6_COMPANY~_ADDRCD_PHONE | 89 | 2 | 0 | 0 |
| F7_GIVEN_ADDRCD_PHONE | 36 | 2 | 3 | 3 |

the rules join 10.9% of the 855 true pairs, with 0 false




## 6 · Candidates

Stage 2 compares only pairs that some generator lists. A key generator pairs records equal on all its keys; a deletion-key generator pairs names one edit apart (each value and its one-character deletions, within a postal area); a sorted window pairs neighbours in sorted order. Each is one SQL join or window over the records table; their pairs are merged, and pairs inside one rule cluster are dropped. Every key here pairs a share of the records that stays flat as the file grows: a field with a fixed pool of values (the phone's last digits, an address shared by a household) is always combined with another. O(N·L + Σ P_g).

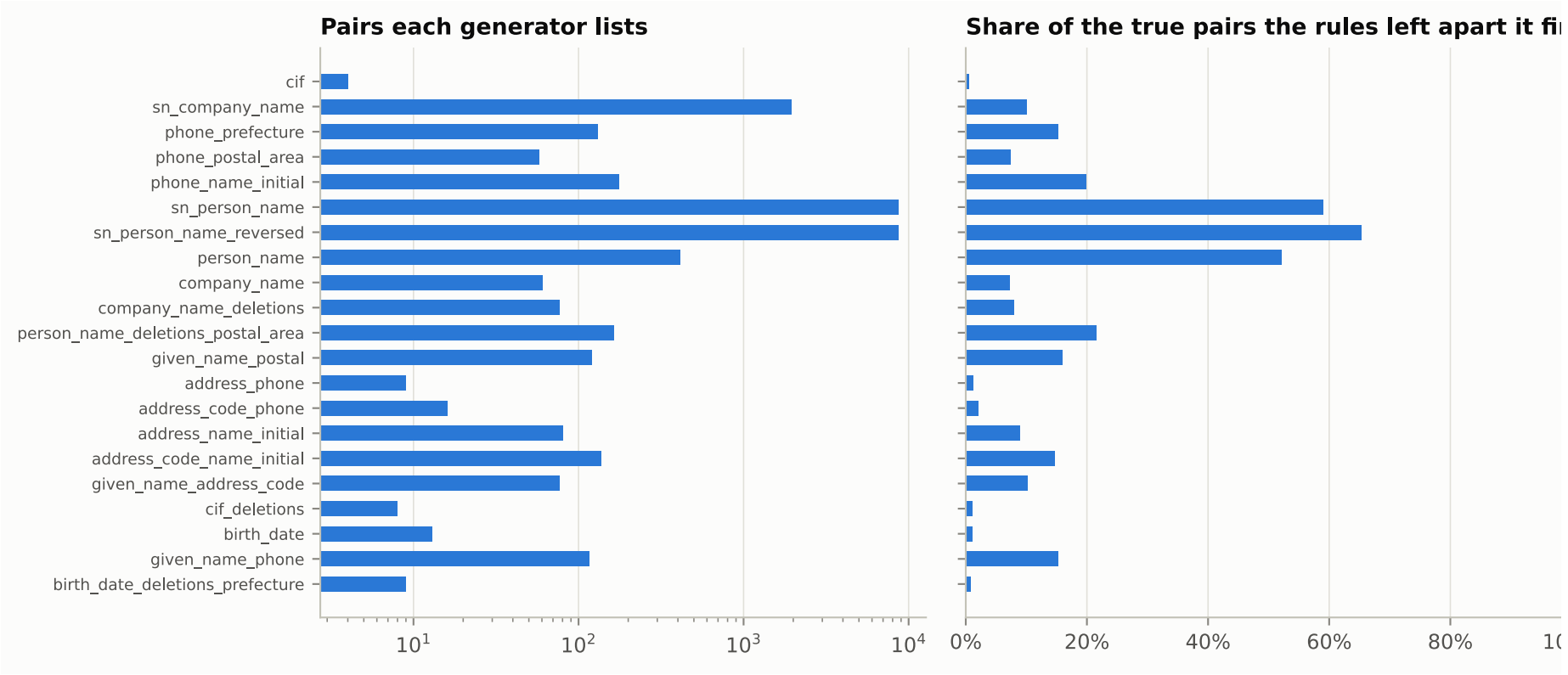
In [6]:
```
report = run.candidates()
done(run)
cands = run.con.sql("SELECT * FROM cands").df().assign(true=same)
targets = missed_pairs(rule_clusters, entity.reindex(rule_clusters.index), UID)  # true pairs the rules left apart
wanted = set(zip(targets[L].astype(str), targets[R].astype(str)))
on_target = np.array([k in wanted for k in zip(cands[L], cands[R])])
report["finds"] = [(found_by(cands, {name}) & on_target).sum() / len(wanted) for name in report["generator"]]
report["different_entity_pairs"] = [(found_by(cands, {name}) & ~cands["true"]).sum() for name in report["generato
r"]]
display(report[["generator", "big_o", "pairs", "finds", "different_entity_pairs", "seconds"]].round(3))
print(f"{len(cands):,} candidate pairs · together they list {on_target.sum() / len(wanted):.1%} of the {len(wante
d):,} true pairs the rules left apart")

fig, (a, b) = plt.subplots(1, 2, figsize=(10, 4.2), sharey=True)
y = np.arange(len(report))[::-1]
a.barh(y, report["pairs"], height=0.6, color=BLUE)
a.set(title="Pairs each generator lists", xscale="log")
b.barh(y, report["finds"], height=0.6, color=BLUE)
b.set(title="Share of the true pairs the rules left apart it finds", xlim=(0, 1))
b.xaxis.set_major_formatter(PercentFormatter(1))
a.set_yticks(y, report["generator"], fontsize=7)
for ax in (a, b):
    ax.grid(axis="y", visible=False)
finish(fig)
```

```
candidates: O(N·L + Σ P_g) = O(3,758·L + 20,977) · 0.12s
```

| | generator | big_o | pairs | finds | different_entity_pairs | seconds |
|---|---|---|---|---|---|---|
| **0** | cif | O(N + P) | 4 | 0.005 | 0 | 0.003 |
| **1** | sn_company_name | O(N log N + N·w) | 1961 | 0.100 | 1885 | 0.002 |
| **2** | phone_prefecture | O(N + P) | 131 | 0.152 | 15 | 0.003 |
| **3** | phone_postal_area | O(N + P) | 58 | 0.073 | 2 | 0.003 |
| **4** | phone_name_initial | O(N + P) | 177 | 0.198 | 26 | 0.003 |
| **5** | sn_person_name | O(N log N + N·w) | 8679 | 0.589 | 8230 | 0.002 |
| **6** | sn_person_name_reversed | O(N log N + N·w) | 8664 | 0.652 | 8167 | 0.002 |
| **7** | person_name | O(N + P) | 414 | 0.521 | 17 | 0.002 |
| **8** | company_name | O(N + P) | 61 | 0.072 | 6 | 0.002 |
| **9** | company_name_deletions | O(N·L + P) | 77 | 0.079 | 17 | 0.002 |
| **10** | person_name_deletions_postal_area | O(N·L + P) | 164 | 0.215 | 0 | 0.004 |
| **11** | given_name_postal | O(N + P) | 121 | 0.159 | 0 | 0.003 |
| **12** | address_phone | O(N + P) | 9 | 0.012 | 0 | 0.003 |
| **13** | address_code_phone | O(N + P) | 16 | 0.021 | 0 | 0.002 |
| **14** | address_name_initial | O(N + P) | 81 | 0.089 | 13 | 0.003 |
| **15** | address_code_name_initial | O(N + P) | 137 | 0.147 | 25 | 0.002 |
| **16** | given_name_address_code | O(N + P) | 77 | 0.101 | 0 | 0.002 |
| **17** | cif_deletions | O(N·L + P) | 8 | 0.010 | 0 | 0.006 |
| **18** | birth_date | O(N + P) | 13 | 0.010 | 5 | 0.002 |
| **19** | given_name_phone | O(N + P) | 116 | 0.152 | 0 | 0.003 |
| **20** | birth_date_deletions_prefecture | O(N·L + P) | 9 | 0.008 | 3 | 0.002 |

18,994 candidate pairs · together they list 87.3% of the 762 true pairs the rules left apart




## 7 · Score

Every candidate pair gets its weight: the prior plus each comparison's bits, with the u section 4 measured on this file. A pair is accepted at the threshold `scripts/choose.py` chose. O(C·K·L²).

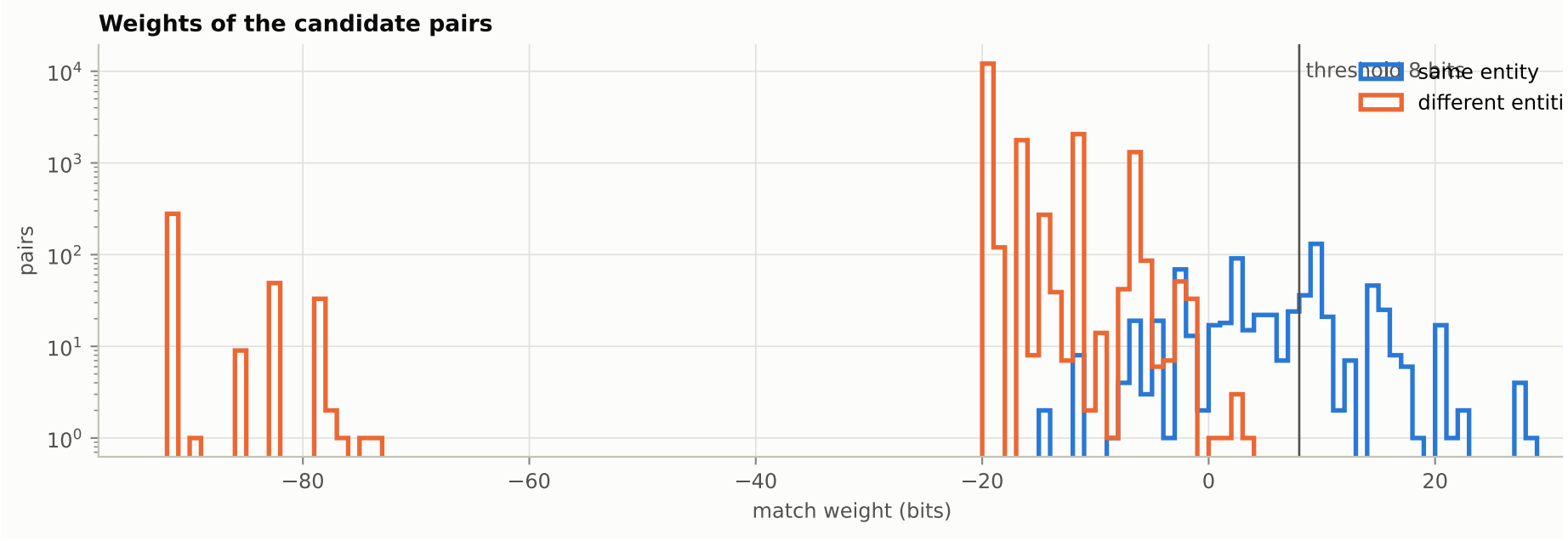
In [7]:

```
run.score()
done(run)
scores = run.con.sql("SELECT * FROM scores").df().assign(same=same)
threshold = run.pipeline.fs.min_match_weight
w = scores["match_weight"]
print(f"{len(scores):,} pairs scored · {(w >= threshold).sum():,} reach {threshold:g} bits, "
      f"{(scores['same'] & (w >= threshold)).sum():,} of them true")

fig, ax = plt.subplots(figsize=(10, 3.4))
```

```
bins = np.arange(math.floor(w.min()), math.ceil(w.max()) + 1)
ax.hist(w[scores["same"]], bins, histtype="step", color=BLUE, linewidth=2, label="same entity")
ax.hist(w[~scores["same"]], bins, histtype="step", color=ORANGE, linewidth=2, label="different entities")
ax.axvline(threshold, color=INK2, linewidth=1)
ax.text(threshold, ax.get_ylim()[1], f" threshold {threshold:g} bits", va="top", color=INK2)
ax.set(title="Weights of the candidate pairs", xlabel="match weight (bits)", ylabel="pairs", yscale="log")
ax.legend(loc="upper right")
finish(fig)
```

```
score: O(C·K·L²) = O(18,994·6·L²) · 0.11s
18,994 pairs scored · 308 reach 8 bits, 308 of them true
```




## 8 · Why a pair matches

A weight is a sum, so it can be read term by term. Two pairs: the accepted true match closest to the threshold whose names differ, and the rejected pair of different entities closest to it. Each bar is one term: the prior, then the bits of the level each comparison landed in.

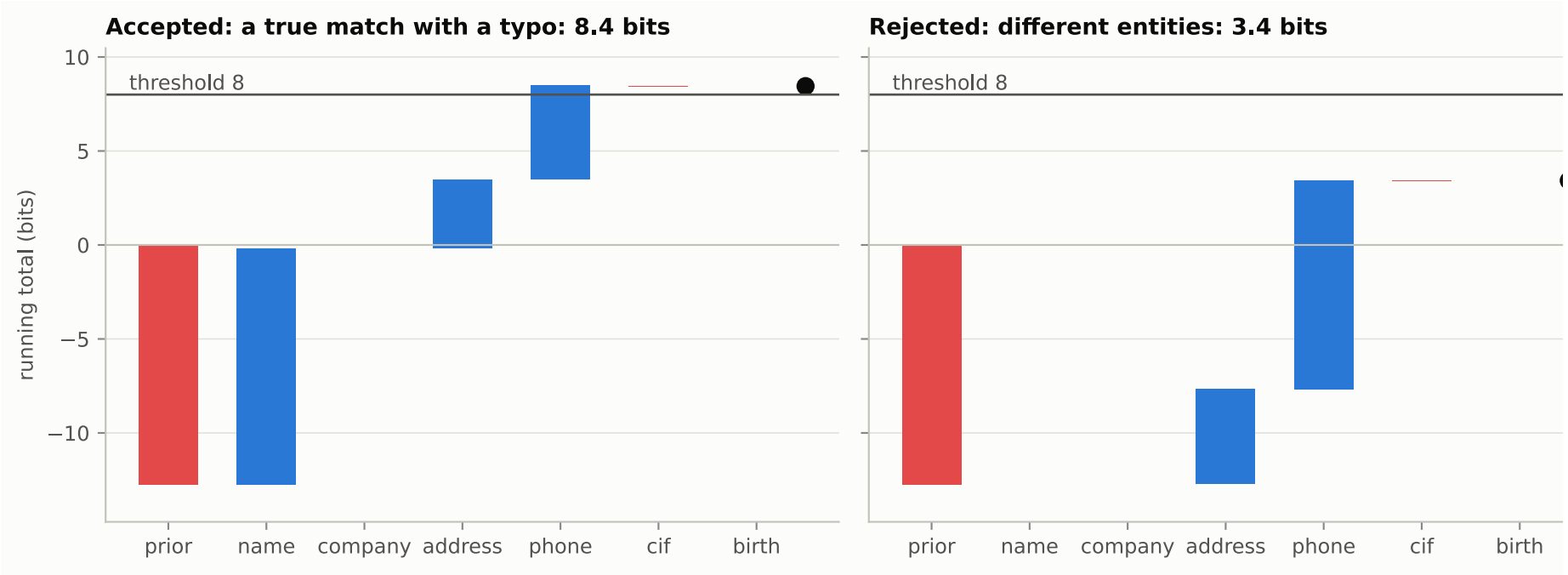
In [8]:
```
show = list(FINAL.explain_show)
filled = list(FINAL.explain_filled)
left, right = std.loc[scores[L], show].set_index(scores.index), std.loc[scores[R], show].set_index(scores.index)
```

```
both_filled = (left[filled].notna() & right[filled].notna()).all(axis=1)
differs = (left[filled[0]] != right[filled[0]]).fillna(False)
accepted = w >= threshold
picked = {"Accepted: a true match with a typo": scores[accepted & scores["same"] & both_filled & differs].nsmalles
t(1, "match_weight"),
          "Rejected: different entities": scores[~accepted & ~scores["same"]].nlargest(1, "match_weight")}

fig, axes = plt.subplots(1, 2, figsize=(10, 3.6), sharey=True)
ends = []
for ax, (title, pair) in zip(axes, picked.items()):
    row = pair.iloc[0]
    display(std.loc[[row[L], row[R]], show])
    bits = waterfall(run.scoring, {c.name: row[f"gamma_{c.name}"] for c in run.pipeline.fs.model.comparisons}, ru
n.n)
    end = bits["bits"].cumsum().to_numpy()
    start = np.concatenate([[0], end[:-1]])
    ends += [*start, *end]
    ax.bar(bits["evidence"], bits["bits"], bottom=start, width=0.6, color=np.where(bits["bits"] >= 0, BLUE, RED))
    ax.plot(len(bits) - 0.5, end[-1], marker="o", markersize=7, color=INK, clip_on=False)
    ax.axhline(threshold, color=INK2, linewidth=1)
    ax.text(-0.4, threshold, f"threshold {threshold:g}", va="bottom", color=INK2)
    ax.axhline(0, color=AXIS, linewidth=0.8)
    ax.set(title=f"{title}: {row['match_weight']:.1f} bits")
    ax.grid(axis="x", visible=False)
axes[0].set(ylabel="running total (bits)", ylim=(min(ends) - 2, max(ends + [threshold]) + 2))
finish(fig)
```

| CONT_ID | full_name | company_name | address | phone | customer_id |
|---|---|---|---|---|---|
| 0000000399 | ヤベケイト | NaN | 東京都昭島市田中町2-12-24 | 4505 | 8766633794722 |
| 0000001035 | ヤベケイド | NaN | 東京都港区西新橋1-20-5 | 4501 | 1909916302348 |

| CONT_ID | full_name | company_name | address | phone | customer_id |
|---|---|---|---|---|---|
| 0000001923 | サンガジシリナオキ | NaN | 茨城県常総市水海道天満町10-8 | 4162 | 1305064375017 |
| 0000002186 | NaN | アサヒアルミサンギョウカブシキガイシャオオサカシテン | 茨城県常総市21-14 | 4162 | 8324699326995 |




## 9 · Threshold

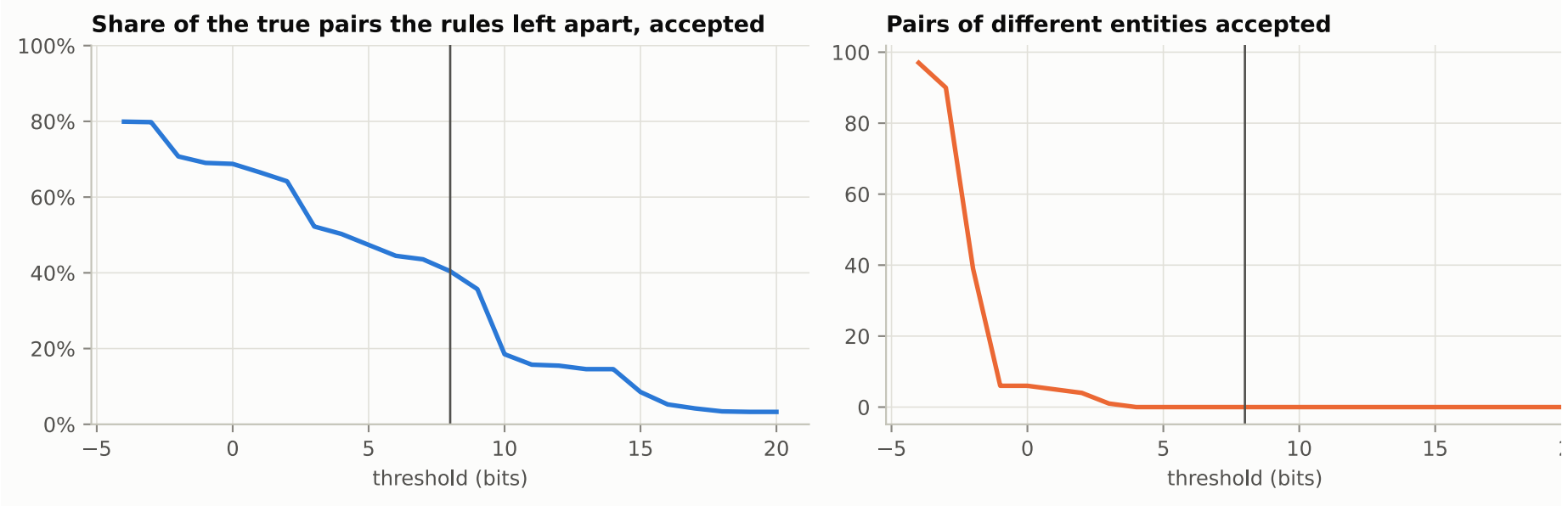
`scripts/choose.py` chose the threshold on cross-validated labelled data: the lowest weight at which no false pair was accepted and the upper bound of the false-merge rate stays under 5%, then one step stricter. A run without labels uses it as it is. Here, with this file's truth: what each weight would accept.

In [9]:
```
with timed(f"O(C·G) = O({len(scores):,} pairs · 25 weights)"):
    grid = np.arange(-4.0, 21.0)
    curve = pd.DataFrame({"weight": grid,
                          "recall": [(scores["same"] & (w >= x)).sum() / len(wanted) for x in grid],
                          "false_pairs": [(~scores["same"] & (w >= x)).sum() for x in grid]})

fig, (a, b) = plt.subplots(1, 2, figsize=(10, 3.2))
a.plot(curve["weight"], curve["recall"], color=BLUE)
a.set(title="Share of the true pairs the rules left apart, accepted", xlabel="threshold (bits)", ylim=(0, 1))
a.yaxis.set_major_formatter(PercentFormatter(1))
b.plot(curve["weight"], curve["false_pairs"], color=ORANGE)
```

```
b.set(title="Pairs of different entities accepted", xlabel="threshold (bits)")
for ax in (a, b):
    ax.axvline(threshold, color=INK2, linewidth=1)
finish(fig)
```

```
O(C·G) = O(18,994 pairs · 25 weights) · 0.00s
```




## 10 · Cluster and evaluate

The accepted pairs and the rules' pairs are clustered together (connected components, in SQL). Gold gives each record its cluster and what resolved it: a rule, or the model. Without labels, the run also says how many of the model's links it expects to be false: the sum of 1 − match probability over them. Then, with the truth: pairwise recall with its 95% interval, false merges, and the upper bound of the false-merge rate. Below: cluster sizes against true entity sizes, and the largest cluster. O(C + E·d + N).

In [10]:

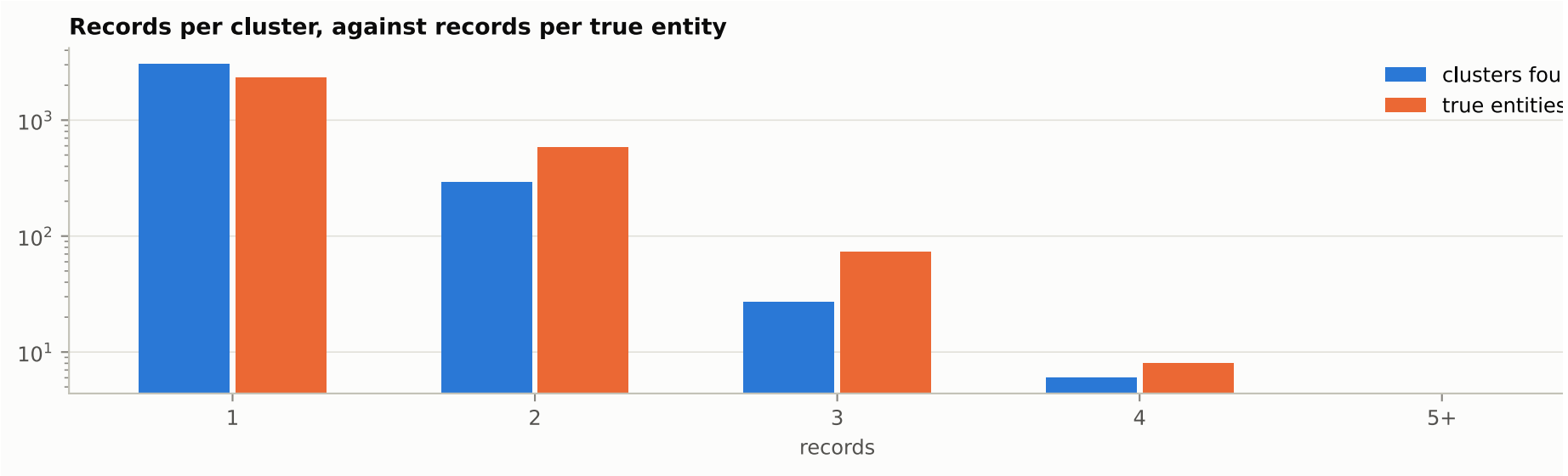
```
run.cluster()
done(run)
shutil.rmtree(DEMO / "run", ignore_errors=True)
run.write(DEMO / "run")
done(run)
gold = pd.read_parquet(DEMO / "run" / "gold.parquet").set_index(UID)
m = cluster_metrics(gold["cluster_id"], entity.reindex(gold.index))
```

```python
linked = next(s for s in run.steps if s["step"] == "cluster")
print(f"model links {linked['accepted']:,} · false as the model expects {linked['expected_false']:.2f} · "
      f"false with the truth {(~same(run.con.sql('SELECT * FROM matches').df())).sum()}")
print(f"recall {m['recall']:.1%} ({m['recall_lo']:.1%}–{m['recall_hi']:.1%}) · false merges {m['FP']} "
      f"(rate at most {m['fp_rate_hi']:.2%}) · precision {m['precision']:.2%}")
display(gold["resolved_by"].fillna("(alone)").value_counts().rename_axis("resolved by").to_frame("records"))

sizes = lambda ids: ids.value_counts().clip(upper=5).value_counts().reindex(range(1, 6), fill_value=0)
fig, ax = plt.subplots(figsize=(10, 3))
x = np.arange(1, 6)
ax.bar(x - 0.16, sizes(gold["cluster_id"]), width=0.3, color=BLUE, label="clusters found")
ax.bar(x + 0.16, sizes(entity), width=0.3, color=ORANGE, label="true entities")
ax.set_xticks(x, ["1", "2", "3", "4", "5+"])
ax.set(title="Records per cluster, against records per true entity", xlabel="records", yscale="log")
ax.grid(axis="x", visible=False)
ax.legend()
finish(fig)
largest = gold["cluster_id"].value_counts().index[0]
display(std.loc[gold.index[gold["cluster_id"] == largest], show].assign(entity=entity))
```

```
cluster: O(C + E·d + N) = O(18,994 + 300·d + 3,758) · 0.02s
write: O(N + C) = O(3,758 + 300) · 0.01s
model links 300 · false as the model expects 0.28 · false with the truth 0
recall 48.0% (44.6%–51.3%) · false merges 0 (rate at most 0.73%) · precision 100.00%
```

| resolved by | records |
|---|---|
| (alone) | 3067 |
| fs | 511 |
| F1_NAME~_ADDRCD_PHONE | 88 |
| X4_PERSON_NAME_ADDRCD_PHONE | 44 |
| F3_NAME~_CIF | 11 |
| F2_NAME~_ADDRESS~_PHONE | 9 |
| F5_COMPANY~_PHONE_POSTAL | 8 |
| F7_GIVEN_ADDRCD_PHONE | 6 |
| F4_NAME_ADDRCD_PHONE~ | 6 |
| X1_CIF_PERSON_NAME | 4 |
| X3_CIF_PHONE | 4 |




| CONT_ID | full_name | company_name | address | phone | customer_id | entity |
|---|---|---|---|---|---|---|
| 0000000640 | コバヤシタツヤ | NaN | 埼玉県ふじみ野市上ノ原1-3-21 | 7543 | 8480021922415 | E001400 |
| 0000000998 | コバヤシタツヤ | NaN | 埼玉県ふじみ野市上ノ原1-3-21 | 7543 | 3057483762514 | E001400 |
| 0000001974 | コバヤシタツヤ | NaN | 埼玉県ふじみ野市上ノ原1-3-21 | 4477 | 8334742939420 | E001400 |
| 0000002491 | コバヤシタツヤ | NaN | 埼玉県さいたま市浦和区駒場2-25-10 | 7543 | 1663636056484 | E001400 |

## 11 · Increment

New records only. An increment joins the batch against all the records, rule by rule and generator by generator, scores the pairs that touch it, and merges into the stored clusters, each existing cluster one node. So its cost grows with the batch, not with the file. Here: a full run on 90% of the records with the model fitted above, then the other 10% as an increment, compared with the full run on all of them in section 10. The increment keeps the stored u, except the exact levels, which it counts again: a comparison's sampled u are measured again only once the drift of its exact level exceeds their sampling error, since a new sample would otherwise only redraw the noise. The steps that follow the data (rules, candidates, cluster) follow the batch.

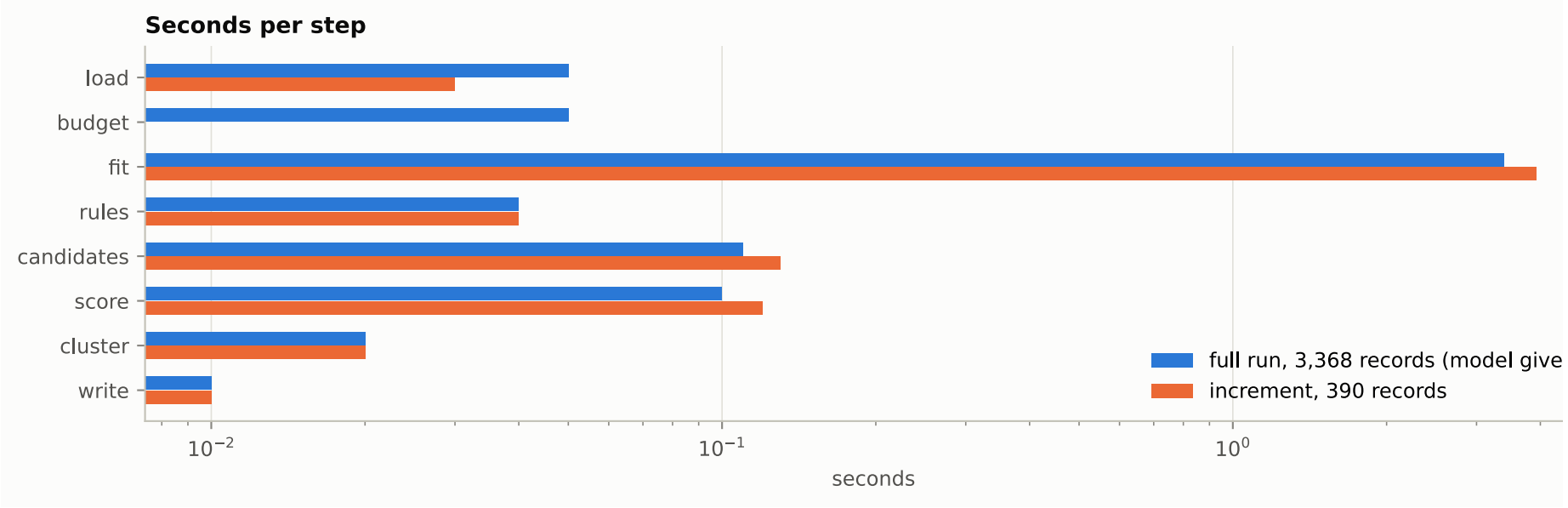
In [11]:
```
batch = np.random.default_rng(0).random(len(g.df)) < 0.1
g.df[~batch].to_parquet(DEMO / "first_90.parquet", index=False)
g.df[batch].to_parquet(DEMO / "new_10.parquet", index=False)
```

```
shutil.rmtree(DEMO / "increment", ignore_errors=True)
base = Run(FINAL, S, model=model).full(DEMO / "first_90.parquet", DEMO / "increment")
inc = Run(FINAL, S).increment(DEMO / "new_10.parquet", DEMO / "increment")
for s in inc.steps:
    print(f"increment {s['step']}: {s['big_o']} = {s['filled']} · {s['seconds']:.2f}s")
for name, b in inc.basis.items():  # what the increment did with u, comparison by comparison
    x = b["last_check"]
    se = x["se_bits"] if x["se_bits"] is not None else float("inf")
    print(f"u of {name}: drifted {x['drift_bits']:.3f} bits, sampling error {se:.3f} → {'sampled again' if x['drif
t_bits'] > se else 'kept'}")

after = pd.read_parquet(DEMO / "increment" / "gold.parquet").set_index(UID)["cluster_id"]
new_ids = set(g.df.loc[batch, UID])
old_in = after[~after.index.isin(new_ids)].unique()
joined = after[after.index.isin(new_ids) & after.isin(old_in)]
truth = predicted_pairs(entity)
by_inc, by_full = predicted_pairs(after), predicted_pairs(gold["cluster_id"])
print(f"{batch.sum():,} new records · {len(joined):,} joined an existing cluster · pairs only in the increment: "
      f"{len((by_inc - by_full) & truth)} true, {len((by_inc - by_full) - truth)} false · only in the full run: "
      f"{len((by_full - by_inc) & truth)} true, {len((by_full - by_inc) - truth)} false")

steps = ["load", "budget", "fit", "rules", "candidates", "score", "cluster", "write"]
seconds = lambda r: pd.Series({s["step"]: s["seconds"] for s in r.steps}).reindex(steps, fill_value=0)
fig, ax = plt.subplots(figsize=(10, 3.2))
y = np.arange(len(steps))[::-1]
ax.barh(y + 0.16, seconds(base), height=0.3, color=BLUE, label=f"full run, {(~batch).sum():,} records (model give
n)")
ax.barh(y - 0.16, seconds(inc), height=0.3, color=ORANGE, label=f"increment, {batch.sum():,} records")
ax.set_yticks(y, steps)
ax.set(title="Seconds per step", xlabel="seconds", xscale="log")
ax.grid(axis="y", visible=False)
ax.legend(loc="lower right")
finish(fig)
```

```
increment load: O(N·L) = O(390·L) · 0.03s
increment fit: O(N + S·K_stale) = O(3,758 + 10,000,000·2) · 3.92s
increment rules: O(|Δ|·Σ b + scan) = O(390 new records) · 0.04s
increment candidates: O(|Δ|·Σ b + scan) = 3,814 pairs touching the batch · 0.13s
increment score: O(C_Δ·K·L²) = O(3,814·6·L²) · 0.12s
increment cluster: O(C_Δ + touched clusters) = 61 accepted · 0.02s
increment write: O(N + C) = O(3,758 + 61) · 0.01s
u of name: drifted 0.063 bits, sampling error 0.017 → sampled again
u of company: drifted 0.072 bits, sampling error 0.126 → kept
u of address: drifted 0.012 bits, sampling error 0.004 → sampled again
u of phone: drifted 0.017 bits, sampling error inf → kept
u of cif: drifted 0.170 bits, sampling error inf → kept
u of birth: drifted 0.336 bits, sampling error inf → kept
390 new records · 73 joined an existing cluster · pairs only in the increment: 6 true, 0 false · only in the full ru
n: 0 true, 0 false
```




## 12 · How the configuration was chosen

`scripts/choose.py` starts from `config/pipeline.toml` and tries the changes in `config/experiments.py` in order, each on top of the choice so far, on labelled synthetic data. A change is adopted by the decision rule: "better" needs a significant paired gain on the selection set; "cheaper" and "safer", no true pair lost; "chosen" is the owner's call, where no false pair may be added. On validation,

all but "chosen" must lose no true pair. Test is scored once, at the end. Its tables are in `reports/choose.md`; read here, not recomputed.

In [12]:
```
text = (S.root / "reports" / "choose.md").read_text(encoding="utf-8")
for stage in ("## Stage 1: rules", "## Stage 2: Fellegi–Sunter"):
    decisions = table_after(text.split(stage, 1)[1], "### Decisions")
    print(stage[3:])
    display(decisions.iloc[:, :8])
print(next(line for line in (S.root / "config" / "final.toml").read_text(encoding="utf-8").splitlines() if line.st
artswith("# Adopted")))
```

```
Stage 1: rules
```

| | change | kind | adopted | train_TP_gains | train_TP_losses | train_p | train_FP_added | train_FP_removed |
|---|---|---|---|---|---|---|---|---|
| 0 | fold_levenshtein | better | True | 17 | 4 | 0.0072 | 0 | 0 |
| 1 | fold_exact | cheaper | False | 0 | 83 | 0.0 | 0 | 0 |
| 2 | f2_address_exact | cheaper | True | 0 | 0 | 1.0 | 0 | 0 |
| 3 | f2_address_prefix | cheaper | True | 0 | 0 | 1.0 | 0 | 0 |
| 4 | optuna_comparators | better | True | 33 | 0 | 0.0 | 0 | 0 |

```
Stage 2: Fellegi–Sunter
```

| | change | kind | adopted | cv_TP_gains | cv_TP_losses | cv_p | cv_FP_added | cv_FP_removed |
|---|---|---|---|---|---|---|---|---|
| 0 | no_term_frequency | better | True | 12 | 0 | 0.0005 | 0 | 0 |
| 1 | exact_keys_only | cheaper | True | 0 | 0 | 1.0 | 0 | 0 |
| 2 | optuna_suggestion | better | True | 1488 | 1 | 0.0 | 0 | 0 |
| 3 | u_from_1e6_pairs | cheaper | False | 1 | 8 | 0.0391 | 0 | 0 |
| 4 | compound_phone_keys | cheaper | True | 0 | 0 | 1.0 | 0 | 0 |
| 5 | sorted_names_for_tfidf | cheaper | True | 1 | 0 | 1.0 | 0 | 0 |
| 6 | name_keys | safer | True | 1 | 0 | 1.0 | 0 | 0 |
| 7 | bounded_address_keys | cheaper | True | 0 | 0 | 1.0 | 0 | 0 |
| 8 | no_sorted_windows | cheaper | False | 0 | 0 | 1.0 | 0 | 0 |
| 9 | bridge_rule_clusters | better | False | 0 | 0 | 1.0 | 0 | 0 |
| 10 | u_from_scored_file | chosen | True | 0 | 60 | 0.0 | 0 | 0 |
| 11 | u_exact_by_counts | chosen | True | 47 | 5 | 0.0 | 0 | 0 |
| 12 | cif_one_digit_off | better | True | 8 | 0 | 0.0078 | 0 | 0 |
| 13 | cif_deletion_key | safer | True | 2 | 0 | 0.5 | 0 | 0 |
| 14 | graded_address | better | True | 124 | 67 | 0.0 | 0 | 0 |
| 15 | birth_date | better | True | 56 | 0 | 0.0 | 0 | 0 |
| 16 | birth_date_key | safer | True | 1 | 0 | 1.0 | 0 | 0 |
| 17 | given_name_phone_key | safer | True | 0 | 0 | 1.0 | 0 | 0 |
| 18 | birth_date_deletion_key | safer | True | 0 | 0 | 1.0 | 0 | 0 |
| 19 | u_near_by_keys | chosen | True | 0 | 2 | 0.5 | 0 | 0 |
| 20 | fit_u_from_1e6_pairs | cheaper | True | 0 | 0 | 1.0 | 0 | 0 |
| 21 | company_without_jw_high | cheaper | True | 1 | 0 | 1.0 | 0 | 0 |
| 22 | name_needed | chosen | True | 0 | 39 | 0.0 | 0 | 0 |

```
# Adopted: rules fold_levenshtein, f2_address_exact, f2_address_prefix, optuna_comparators; Fellegi–Sunter no_term_frequency, exact_keys_only, optuna_suggestion, compound_phone_keys, sorted_names_for_tfidf, name_keys, bounded_address_keys, u_from_scored_file, u_exact_by_counts, cif_one_digit_off, cif_deletion_key, graded_address, birth_date, birth_date_key, given_name_phone_key, birth_date_deletion_key, u_near_by_keys, fit_u_from_1e6_pairs, company_without_jw_high, name_needed.
```

## 13 · Cost as the file grows

Every step's Big-O, for this full run and for the increment, with the numbers filled in. A full run is $O(N{\cdot}L + \Sigma\ P + C{\cdot}K{\cdot}L^2 + E{\cdot}d)$, plus the fit and the u sample, whose sizes are fixed. When every key's pairs per record stay flat, that is linear in N. An increment of Δ records costs

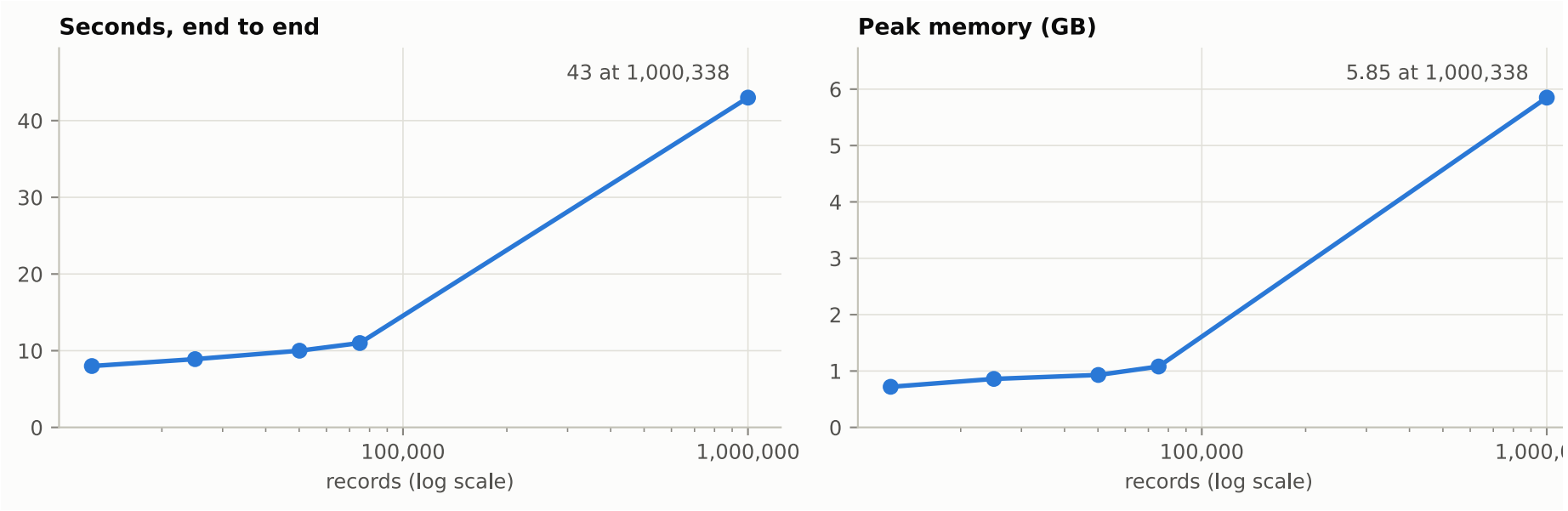
O(|Δ|·(L + Σ b) + C_Δ·K·L²) plus scans of the stored records. Below, measured by `scripts/benchmark.py` (`reports/benchmark.md`): each file resolved in a process of its own, up to a million records.

In [13]:
```
full_steps = pd.DataFrame(run.steps)[["step", "big_o", "filled", "seconds", "peak_memory_gb"]]
inc_steps = pd.DataFrame(inc.steps)[["step", "big_o", "filled", "seconds"]]
display(full_steps, inc_steps)

curve = table_after((S.root / "reports" / "benchmark.md").read_text(encoding="utf-8"), "## Cost")
curve = curve[pd.to_numeric(curve["seconds"], errors="coerce").notna()].astype({"records": int, "seconds": float,
"peak_memory_gb": float})
fig, (a, b) = plt.subplots(1, 2, figsize=(10, 3.2))
for ax, col, title in ((a, "seconds", "Seconds, end to end"), (b, "peak_memory_gb", "Peak memory (GB)")):
    ax.plot(curve["records"], curve[col], color=BLUE, marker="o", markersize=6)
    ax.set(title=title, xlabel="records (log scale)", xscale="log", ylim=(0, curve[col].max() * 1.15))
    ax.xaxis.set_major_formatter(FuncFormatter(lambda x, _: f"{x:,.0f}"))
    last = curve.iloc[-1]
    ax.annotate(f"{last[col]:g} at {last['records']:,}", (last["records"], last[col]), textcoords="offset points",
                xytext=(-8, 8), ha="right", color=INK2)
finish(fig)
```

| | step | big_o | filled | seconds | peak_memory_gb |
|---|---|---|---|---|---|
| 0 | load | O(N·L) | O(3,758·L) | 0.09 | 0.46 |
| 1 | fit | O(n + S·K + P_em·K) + O(N + S·K) | n = 3,758; u on 3,758 records | 5.38 | 0.72 |
| 2 | budget | O(N) | O(3,758) per rule and generator | 0.06 | 0.73 |
| 3 | rules | O(N + Σ P_r·L²) + O(E·d + N) | O(3,758 + 91 pairs) | 0.04 | 0.74 |
| 4 | candidates | O(N·L + Σ P_g) | O(3,758·L + 20,977) | 0.12 | 0.75 |
| 5 | score | O(C·K·L²) | O(18,994·6·L²) | 0.11 | 0.75 |
| 6 | cluster | O(C + E·d + N) | O(18,994 + 300·d + 3,758) | 0.02 | 0.76 |
| 7 | write | O(N + C) | O(3,758 + 300) | 0.01 | 0.76 |

| | step | big_o | filled | seconds |
|---|---|---|---|---|
| 0 | load | O(N·L) | O(390·L) | 0.03 |
| 1 | fit | O(N + S·K_stale) | O(3,758 + 10,000,000·2) | 3.92 |
| 2 | rules | O(\|Δ\|·Σ b + scan) | O(390 new records) | 0.04 |
| 3 | candidates | O(\|Δ\|·Σ b + scan) | 3,814 pairs touching the batch | 0.13 |
| 4 | score | O(C_Δ·K·L²) | O(3,814·6·L²) | 0.12 |
| 5 | cluster | O(C_Δ + touched clusters) | 61 accepted | 0.02 |
| 6 | write | O(N + C) | O(3,758 + 61) | 0.01 |


Seconds, end to end
43 at 1,000,338
40
30
20
10
0
100,000
1,000,000
records (log scale)
Peak memory (GB)
5.85 at 1,000,338
6
5
4
3
2
1
0
100,000
records (log scale)